

Multiple Choice

1. **Answer:** E

Options: A) Constant energy levels across all trophic levels; B) Increase in nutrient availability at each level; C) Decrease in carbon dioxide concentration; D) Increase in biomass at each successive level; E) Great decrease (90%) in available energy at each level

Note: Correct option: E - Great decrease (90%) in available energy at each level

2. **Answer:** A

Options: A) Sponge; B) Protist; C) Clam; D) Starfish; E) Coral

Note: Correct option: A - Sponge

3. **Answer:** A

Options: A) an individual's genome; B) a population's genetic variation; C) a species' genetic mutation; D) an individual's genetic makeup; E) a population's gene frequency

Note: Correct option: A - an individual's genome

4. **Answer:** C

Options: A) parenchyma; B) phloem; C) endoderm; D) pericycle; E) cortex

Note: Correct option: C - endoderm

5. **Answer:** A

Options: A) Oncogenes are found only in viruses; B) Oncogenes prevent cancer cells; C) Oncogenes are a type of immune cell; D) Oncogenes detoxify harmful substances in the cell; E) Oncogenes slow down cell division

Note: Correct option: A - Oncogenes are found only in viruses

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Yes, viruses can be a cause of cancer'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Mate attraction, species recognition, defense, and territorial claim'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1/53'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'amnion'.

Long Form Response

11. **Answer:** Short loops of Henle to maximize water secretion. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** When we swallow, the larynx moves up so that the glottis is closed by the epiglottis and the food or liquid passes into the esophagus behind the trachea. Swallowing is a complex reflex initiated when pressure receptors in the wall of the pharynx are stimulated.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key